

Product improvement explanation

66% of the audience chooses the more sustainable packaging (redesign).

The sterole dosage in the old design was only verbal and inspecific. To elevate it, it was chosen to be structural by setting the average daily dosage. This has reduced the misunderstanding of the claim and made it more specific.

Greenwashing elements were eliminated and instead a sterol chemical structure was put to enhance the clarity of the product ingredients.

The change between the original packaging and the redesign worked to a great extent because where to dispose was clearer using the icons, and the redesign also tackled the “motivation problem” according to the COMbi analysis that was executed. This was done by making a lottery and explanation was implemented through a QR code. Therefore, it could communicate better where and how it can be disposed.

A total of 12 participants evaluated the redesigned packaging using a 5-point scale in comparison to the original packaging. The envisioned behaviour was that users would:

- 1-perceive the package as attractive (or standing out)
- 2-easily understand its ingredients information
- 3-clearly know how to dispose of it



The redesign questions covered 5 aspects with their hypothesis

Visual appeal	The packaging is redesigned with improved clearer visual communication with icons and sustainable (structural, visual and verbal) cues than the old packaging, users will perceive the package as visually appealing (average rating ≥ 4).
Motivation to dispose	If the redesign improves the costumer behaviour towards the motivation to separate the materials of the packaging and dispose it .(average rating ≥ 3.5). This is with keeping the colour palette and the theme of the product the same
Disposal clarity	If the packaging integrates disposal instructions clearly users will clearly understand how to dispose of the packaging (average rating ≥ 4). It is also expected to find alot of discrete difference between the numbers during testing because some people come from different environment that explains the way a package should be disposed differently.
Instruction transparency	If disposal instructions are communicated through icons and text, users will perceive the instructions as clear and understandable (average rating ≥ 4).
Ingrediant transparency	The ingredient information is communicated transparently, users will perceive the packaging as transparent about its contents (average rating ≥ 4).

Data analysis

and the test plan

- 1 Poor
- 2 Fair
- 3 Average
- 4 Good
- 5 Excellent

Question (criteria)	Average score	Percentage of maximum score
To what extent does the package visually appealing	4.7	94%
The package motivates the consumer to separate the material before disposing	4.6	92%
Its is obvious how to dispose the package	3.6	72%
The packaging has clear instructions on how to dispose	3.3	66%
The packaging has transparent informations about its ingredients	4.5	90%

Received feedback (observation)

The general feedback that was received that the disposal place could be improved because the redesign draft did not directly state where it should be disposed. It was because during designing the main focus was to design something that would motivate the consumers to separate the material and dispose it but it was not clear where to dispose it.



Redesign



Original

Key findings

Visual appeal (4.7 / 94%)

The packaging is perceived as highly appealing. This indicates that the design aesthetics, the used softer colours, and layout successfully attracted people's attention.

Motivation to dispose (4.6 / 92%)

The packaging significantly showed the motivational behaviour to dispose the package when separating the layers and scrabbing the residuals.

Disposal clarity (3.6 / 72%)

The consumers somewhat understand how to dispose of the packaging, but the clarity is not optimal since it was not the main focus for the packaging.

Disposal instructions (3.3 / 66%)

Instructions are perceived as only partially clear, suggesting that icons for the demonstration part would need improvement because a big part did not understand what the icons were trying to communicate

Ingredient transparency (4.5 / 90%)

Users perceive ingredient information as transparent and accessible due to explaining the sterol ingredient and using its leaf and the chemical structure for transparency

Reflection on the Hypothesis

H1 – Visual Appeal→Accepted.

Average of 4.7 indicates that the packaging redesign successfully improves visual attractiveness. Participants found the design engaging and appealing.

H2 – Motivation to dispose→ Accepted

4.6 suggests that the packaging redesign successfully improved the motivation for the participants to separate the materials

H3 – Disposal Clarity→ Partially accepted.

3.6 means that users understand disposal methods but still experience uncertainty. According to the feedback, it was due to the unclear recycling symbols

H4 – Instruction Transparency→Not acceptable.

3.3 means that disposal instructions are insufficiently clear. It does not have a PMD icon

H5 – Ingredient Transparency→Acceptable.

The high score 4.5 confirms that the packaging successfully communicates ingredient information clearly

Evaluation of behavioural change

The envisioned behaviour was for users to recognize the product, understand its sustainability according to the cues, and dispose of it correctly.

The design partially achieves the intended behavioural outcome in terms of disposing the package into the PMD disposal bin, most of them will just throw it according to their knowledge but there is no clear instruction that says where it should be thrown.

The design however did achieve the behavioural outcome from the aspect of separating the material inside the butter box, scrapping the residual to win the lottery, which reflects the solution for the main problem

Perceived Sustainability Change

The redesigned packaging appears to improve the perceived sustainability score of the product, particularly through the transparent ingredient communication (4.5) and the motivation to dispose (4.6). However, sustainability communication could be further strengthened by making recycling instructions clearer and more visible by using different icons than the used ones.

Design improvements for Future Packaging

1. Improve disposal communication

Sustainability instructions for disposing should be simplified and visually dominant, this can happen by balancing where to dispose the packaging and the motivation to throw it (the lottery). The visual cues should be enhanced by using icons that are more universally known so a wide variety of people could understand what the instruction is trying to communicate through that icon.

2. Maintain transparency

The sterol ingredient transparency was highly appreciated by users and should remain a key design strength.

